

Noise Growth and Parameter Selection in Fully Homomorphic Encryption

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Recap: What are the parameters of an FHE scheme?

Almost all FHE schemes used today are based on the Learning with Errors (LWE) problem (or its algebraic variant):

Definition (*Search* LWE Problem (Regev))

Given $\mathbf{b} \in \mathbb{Z}_q^m$ and $A \in (\mathbb{Z}_q)^{m \times n}$, **find** an *unknown* vector $\mathbf{s} \in \mathbb{Z}_q^n$ such that

$$A\mathbf{s} + \mathbf{e} = \mathbf{b} \bmod q$$

where $\mathbf{e} \in \mathbb{Z}_q^m$ is *small* random error.

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- $\mathbf{s}$ follows the distribution χ_s with standard deviation σ_s .
- $\mathbf{e}$ follows the distribution χ_e with standard deviation σ_e .
- Modulus q and the LWE dimension n .

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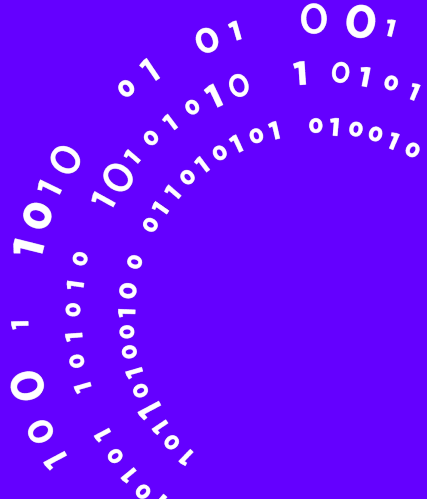
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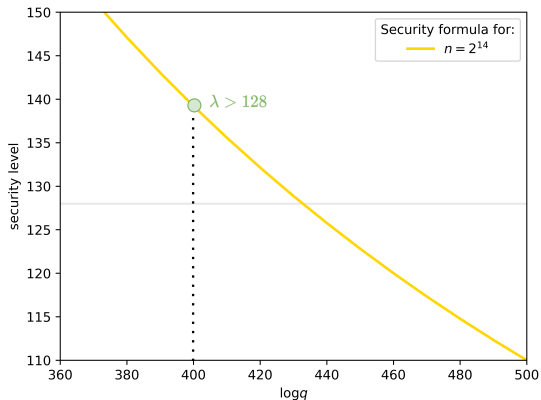
Relation between parameters



What a complex life!

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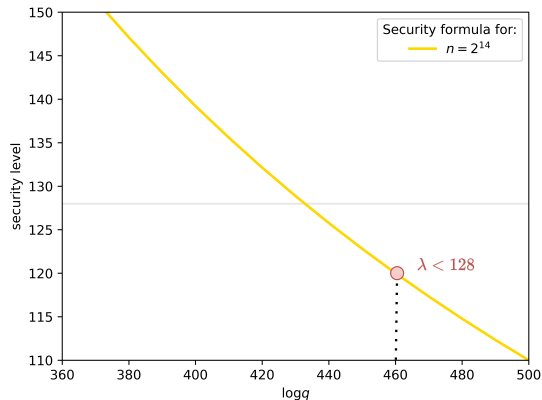
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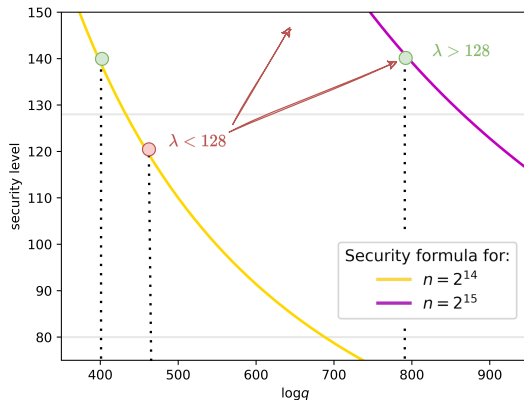
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operations $\uparrow$ e $\uparrow$ q $\uparrow$ λ $\downarrow$



What a complex life!

- 1 To guarantee **correctness**, we need a large enough modulus q .
- 2 **Security problem**: larger q decreases the security.
- 3 **Efficiency problem**: To increase the security level λ again, we need a larger dimension n .



(Some of the FHE) Problems

- All the parameters are related to each other.



- 1 **Cascading Effect**: Altering one parameter triggers a ripple effect, influencing other parameters throughout the scheme.
 - 2 **Trade-off** between security (small q) and error margin (large q): very complicated to find a **optimal set** (namely, secure and efficient) of parameters.
- **Powers-of-two are spars**: the jump between two consecutive powers-of-two is *big*.

What can we do?

1. Compute e properly
2. Find a **simple** formula that links security and the other parameters
3. Provide an **automatic parameters generator** for *any* circuit
4. Find a **good** $\Phi_m(x)$ which has not a power-of-2 degree

Small parenthesis on non-power-of-2 polynomials

Find a **good** $\Phi_m(x)$ which has not a power-of-2 degree

We have to take care of:

- How **reduction modulo** the quotient cyclotomic polynomial $\Phi_m(x)$ impacts
 - the computation of polynomial products;
 - the error bounds.
- “Fast” algorithms for the Number Theoretic Transform (NTT);

Small parenthesis on non-power-of-2 polynomials

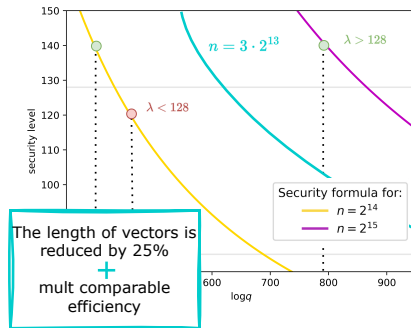
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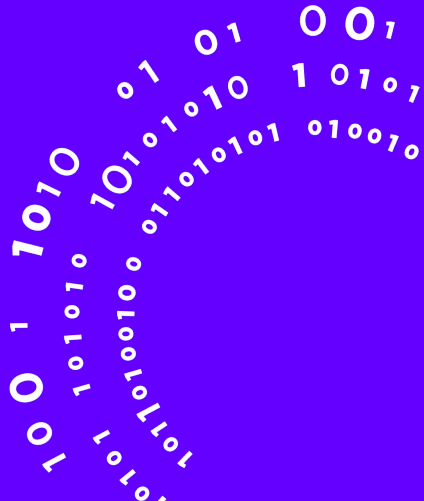
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One possibility $m = 2^i 3^j \implies n = m/3$ and

$$\Phi_m(x) = x^n - x^{n/2} + 1$$



Worst-Case vs Average-Noise analysis



Noise analysis

We want e as small as possible.

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Compute $e = ||\nu||$ properly



Current Approaches for studying the error

Ciphertexts are random polynomials $c(x) \in \mathcal{R}_q$.

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$\implies$ The **canonical norm** is $\|a\|^{can} = \max_{j \in \mathbb{Z}_m^*} |a(\xi^j)|$. It is the norm of the canonical embedding:

$$\begin{array}{ccc} \sigma : \mathcal{R} = \mathbb{Z}[x] / \langle \Phi_m(x) \rangle & \rightarrow & \mathbb{C}^{\phi(m)} \\ a & \mapsto & (a(\xi^j))_j \end{array}$$

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Theorem (Di Giusto, -)

Let $a \in \mathcal{R}_q$ be a random polynomial whose coefficients are **identically distributed**, have **mean zero** and variance V_a , then the distribution of $a(\xi)$ is well approximated by a centered **Gaussian distribution** with variance $\phi(m)V_a$.

Recall

Variance is a measure of dispersion, meaning it is a measure of how far a set of numbers is spread out from their average value.

The **standard deviation** is the square root of the variance.

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★ $\|a\|_\infty \leq D\sqrt{2V_a}$

★ $\|ab\|_\infty = \frac{\text{explosion}}{\text{bomb}} < \|ab\|^{can}$

★ $\|a\|^{can} \leq \phi(m)\|a\|_\infty$

★ $\|a\|_\infty \leq c_m\|a\|^{can}$
for the **ring expansion factor** c_m
(if $m = 2^\kappa \implies c_m = 1$)

Average-case controversy



The average-case provides estimates that are **closer to experimental** observations.

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It relies on two assumptions that have recently been called into question.

Most existing approaches model the coefficients of the critical quantity as:

- Independent.
- Gaussian distributed.

A Critique on Average-Case Noise Analysis in RLWE-Based Homomorphic Encryption

Authors:  Mingyu Gao,  Hongren Zheng [Authors Info & Claims](#)

WAHC '25: Proceedings of the 13th Workshop on Encrypted Computing & Applied Homomorphic Computing
Pages 26 - 37 • <https://doi.org/10.1145/3733811.3767312>

Published: 13 October 2025 [Publication History](#)

 Check for updates

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Abstract  AI Summary

Homomorphic encryption schemes based on the Ring-Learning-with-Errors problem require

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Who is right?

BGV scheme - The three basic algorithms

■ Key(λ, L):

- ★ Sample $s \leftarrow \chi_s, a \leftarrow \mathcal{U}_{q_L}$ and $e \leftarrow \chi_e$ in $\mathbb{R}_{q_L}$,
- ★ Output the secret key $\mathbf{sk} = s$ and the public key $\mathbf{pk} = (b, a) = [(-a \cdot s + te, a)]_{q_L}$.

■ Enc $_{\mathbf{pk}}(m)$:

- ★ Receive a plaintext $m \in \mathbb{R}_t$ and the public key $\mathbf{pk} = (b, a)$;
- ★ Sample $u \leftarrow \chi_s^n, e_0, e_1 \leftarrow \chi_e^n$;
- ★ Output $\mathbf{c} = (\mathbf{c}, L, \nu)$ where the ciphertext is

$$\mathbf{c} = (c_0, c_1) = [(b \cdot u + te_0 + m, a \cdot u + te_1)]_{q_L}.$$

$\mathbf{c}$ is called **extended ciphertext**.

■ Dec $_{\mathbf{sk}}(\mathbf{c})$:

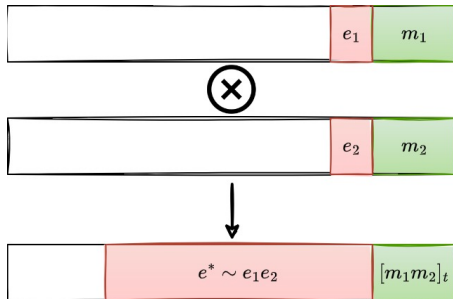
- ★ Receive the secret key $\mathbf{sk}$ and the extended ciphertext $\mathbf{c} = (\mathbf{c}, \ell, \nu)$;
- ★ Output

$$m = [[c_0 + c_1 \cdot s]_{q_\ell}]_t.$$

Note that

$$\nu = [c_0 + c_1 \cdot s]_{q_\ell} = [(-a \cdot s + te) \cdot u + te_0 + m + (a \cdot u + te_1) \cdot s]_{q_\ell} = [m + t(e \cdot u + e_1 \cdot s + e_0)]_{q_\ell} \mathbf{9}$$

The BGV Example on average-case noise analysis



$$\nu_1 = m_1 + te_0 + tu \cdot e + te_1 \cdot s$$



$$\nu_2 = m_2 + te'_0 + tu' \cdot e + te'_1 \cdot s$$



$$\nu_1 \nu_2 = m_1 m_2 + (\dots) + (\dots)e + (\dots)s + (\dots)es + (\dots)e^2 + (\dots)s^2$$

The noise always has this form

$$\nu = \sum_{\iota} a_{\iota} s^{\iota} = \sum_{\iota} \sum_{\mu} b_{\mu, \iota} e^{\mu} s^{\iota}$$

First Assumption: Independency

Are the noises independent?

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$\Rightarrow$ Errors share powers of s and e

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Let V, V' be the variances of the noise coefficients of two ciphertexts. Then, the variance of the noise coefficient obtained by a multiplication of these two ciphertexts is

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Given $\nu_1 = \sum_{\iota} a_{\iota} s^{\iota}$, $\nu_2 = \sum_{\iota} a'_{\iota} s^{\iota}$, then

$$V((a_{\iota_1} s^{\iota_1} a'_{\iota_2} s^{\iota_2})|_i) \leq n V((a_{\iota_1} s^{\iota_1})|_i) V((a'_{\iota_2} s^{\iota_2})|_i) F(\iota_1, \iota_2) F(\mu_1, \mu_2),$$

where μ_1, μ_2 represent, respectively, the highest power of e in $a_{\iota_1}, a'_{\iota_2}$.

$$F(\alpha, \beta) = \frac{(\alpha + \beta)!}{\alpha! \beta!}$$

Example – 1 multiplication

$$\nu_{\text{clean}} = m + t(e \cdot u + e_1 \cdot s + e_0) = \sum_{\mu=0}^1 \sum_{\iota=0}^1 b_{\mu,\iota} e^{\mu} s^{\iota} = \sum_{\iota=0}^1 a_{\iota} s^{\iota} \implies V_0$$

where $a_0 = m + te \cdot u + te_0$ and $a_1 = te_1$.

Multiplying $\mathbf{c}' = \text{Enc}(m')$ and $\mathbf{c}'' = \text{Enc}(m'')$ we get

$$\nu_{\text{mul}} = \alpha_0 + \alpha_1 s + \alpha_2 s^2 = a'_0 a''_0 + (a'_0 a''_1 + a''_0 a'_1) s + a'_1 a''_1 s^2$$

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$$\begin{aligned} V(\nu_{\text{mul}}) \leq & nV(a'_0|i)V(a''_0|i)F(1,1)F(0,0) + nV(a'_0|i)V(a''_1s|i)F(1,0)F(0,1) + \\ & + nV(a''_0|i)V(a'_1s|i)F(0,1)F(1,0) + nV(a'_1s|i)V(a''_1s|i)F(0,0)F(1,1) \end{aligned}$$

$$\implies V(\nu_{\text{mul}}) \leq 2nV(a'_0|i)V(a''_0|i) + nV(a'_0|i)V(a''_1s|i) + nV(a''_0|i)V(a'_1s|i) + 2nV(a'_1s)V(a''_1s)$$

$$\implies V(a_0|i) \approx V((a_1s)|i) \approx \frac{V_0}{2} \implies V(\nu_{\text{mul}}) \leq 6n \frac{V_0^2}{4} = \frac{3}{2} n V_0^2$$

Experimental Validation

Key Idea: Let V_1, V_2 be the noise variances of two ciphertexts.

variance of the product $\neq$ product of variances

$\Rightarrow$ Dependencies introduces a **correction factor** $\times nV_1V_2 \longrightarrow \checkmark nV_1V_2 \cdot F_sF_e$

	Encryption		1 Multiplication		6 Multiplications	
n	<i>our</i>	<i>exp</i>	<i>our</i>	<i>exp</i>	<i>our</i>	<i>exp</i>
2^{13}	48.76	48.76	95.68	95.65	95.71	95.25
2^{14}	49.76	49.76	98.68	97.62	98.71	97.63
2^{15}	50.76	50.72	101.68	101.62	101.71	101.59

Estimated variances (in $\log_2$) of the error coefficients obtained using *our* = our estimates, and *exp* = experimental results via OpenFHE.

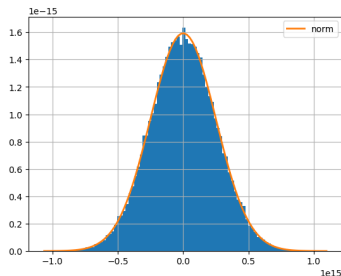
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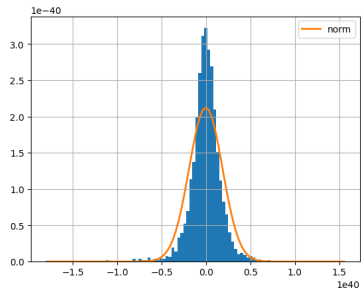
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Short Answer: **it depends!**



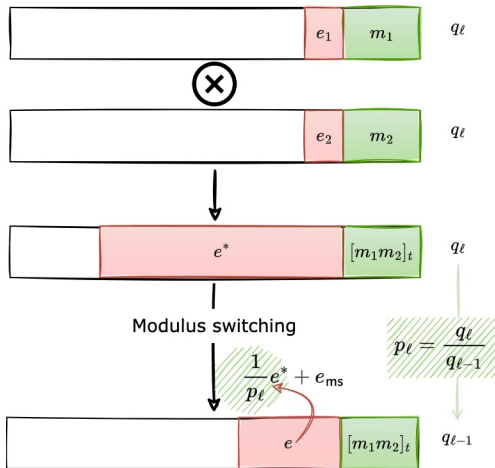
Error coefficients distribution after three multiplications *with modulus switching*: asymmetry 0.00 and kurtosis 2.99.



Error coefficients distribution after three multiplications *without modulus switching*: asymmetry -0.52 and kurtosis 14.38.

Modulus Switch

Modulus switch consists in **reducing the error** by reducing the ciphertext modulus.
We add a **small correction term δ** to $\mathbf{c}$



$$\mathbf{c}' = \frac{1}{p_\ell} (\mathbf{c} + \delta) \mod q_{\ell-1}$$

■ **Noise** after the modulus switch:

$$\nu^* \implies \frac{\nu^*}{p_\ell} + \delta \cdot s$$

■ **Variance** after the modulus switch:

$$V \implies \frac{V}{p_\ell^2} + V_{ms}$$

The term V_{ms} is the price of the correction: we divide the old variance, and pay a new fixed one.

Our moduli choice

ERROR AFTER MODULUS SWITCH

$$\nu = \frac{\nu^*}{p_\ell} + \underbrace{\delta \cdot \mathbf{s}}$$

We proved it is
Gaussian distributed

VARIANCE AFTER MODULUS SWITCH

$$\frac{V}{p_\ell^2} + V_{\text{ms}}$$

Thus, by choosing p_ℓ large enough
with coefficients variance

$$\begin{aligned}\nu &\approx \delta \cdot \mathbf{s} \\ &\approx V_{\text{ms}}\end{aligned}$$

Specifically, we required

$$p_\ell > \sqrt{\frac{V}{\alpha \cdot V_{\text{ms}}}}$$

where we choose $\alpha = \frac{1}{100}$.

By choosing p_ℓ large enough, gaussianity is guaranteed.

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Let V, V' be the variances of the noise coefficients of two ciphertexts. Then, the variance of the noise coefficient obtained by a multiplication of these two ciphertext is

$$V_{\text{mul}} \leq n F_s F_e V V'$$

where F_s and F_e depending on the number of multiplications previously performed in the circuit.

$\Rightarrow$ NO mod switch then $V_{\text{mul}} \approx n F_s F_e V V'$ BUT $\|\nu\|_{\infty} \not\leq D \sqrt{2 V_{\text{mul}}}$

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Let V, V' be the variances of the noise coefficients of two ciphertexts. Then, the variance of the noise coefficient obtained by a multiplication of these two ciphertext is

$$V_{\text{mul}} \leq n \cdot 2 \cdot 1 \cdot V V'$$

*where F_s and F_e **not** depending on the number of multiplications previously performed in the circuit.*

$\implies$ with mod switch then $V_{\text{mul}} \approx 2nV V' \approx 2nV_{\text{ms}}^2$ AND $\|\nu\|_{\infty} \leq D\sqrt{2V_{\text{mul}}}$

Why is it better than worst-case?

$$\|\nu_{\text{mul}}\|^{\text{can}} \leq \|\nu\|^{\text{can}} \|\nu'\|^{\text{can}} \leq D^2 n \sqrt{VV'} \quad \text{VS} \quad \|\nu_{\text{mul}}\|_{\infty} \leq D \sqrt{2V_{\text{mul}}} = D \sqrt{2n2VV'}$$

3 multiplications
(BFV example)

n	can	our	exp
	$t = 2^{16} + 1$		
2^{13}	45.0	62.5	63.6
2^{14}	41.0	59.1	60.1
2^{15}	37.0	55.6	56.4

bits missing for wrap around

n	2^{12}	2^{13}	2^{14}	2^{15}
can	132.6	136.6	140.6	144.6
our	114.5	117.9	121.4	125.0

14%
 $\log q$

Biasioli, Marcolla, Calderini, Mono – *Improving and Automating BFV Parameters Selection: An Average-Case Approach* (2023)

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HElib

$$\log_2(p_{\ell}) \approx 59$$

n	2^{12}	2^{13}	2^{14}	2^{15}
$\log_2(p_{\ell})$	29.55	30.55	31.55	32.55

OpenFHE

$$\log_2(q)$$

n	2^{12}	2^{13}	2^{14}	2^{15}
OpenFHE	147.3	151.8	156.3	161.6
our	121.0	124.5	128.0	131.5

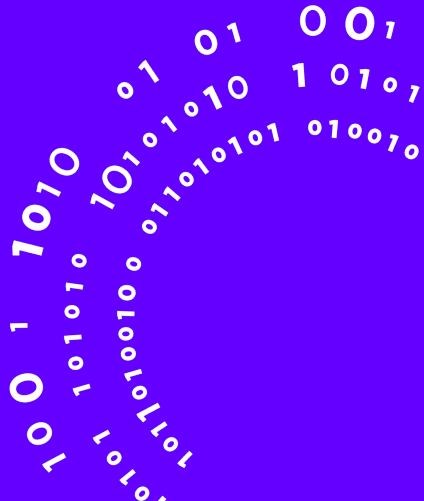
(a) Circuit of depth 3.

n	2^{12}	2^{13}	2^{14}	2^{15}
OpenFHE	249.3	256.8	264.3	272.6
our	210.3	216.8	223.3	229.8

(b) Circuit of depth 6.

Biasioli, Marcolla, Murru, Urani – *Accurate BGV parameters selection: Accounting for secret and public key dependencies in average-case analysis* (2025)

Security of (R)LWE-based schemes



Security of schemes based on (R)LWE problems

All FHE schemes (used now) are based on the LWE problem or its ring variant



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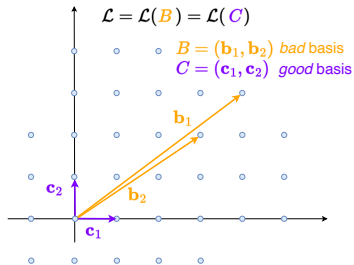


- The **security** of FHE schemes depends on the **intractability of the LWE** problem.
- **Attacks on FHE** schemes are based on finding efficient **algorithms to solve lattice problems**.

The central part of these algorithms is based on a **lattice reduction**: starting from a *bad* lattice basis, find a *better* basis.



The BKZ- β (Block-Korkine-Zolotarev) algorithm works by calling multiple times an algorithm for SVP on sublattices of dimension β .



The star β



- 1** In any lattice-based attack, the **main role** is played by the lattice reduction parameter β
 $\implies$ determine the *optimal* $\beta = f(n, q, \sigma_e, \sigma_s)$.

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$$T_{\text{BKZ}}(\beta, d) = 8d \cdot 2^{0.292\beta + 16.4} \implies \lambda = 0.292\beta + \log_2(8d) + 16.4,$$

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- 3 The previous expression can be reverted for any desired parameter, e.g.

$$n = g(\lambda, q, \sigma_e, \sigma_s).$$

How are FHE Parameters Selected Today?

1. First Approach – Lattice Estimator

The [Lattice Estimator](https://github.com/malb/lattice-estimator) (<https://github.com/malb/lattice-estimator>) [1] is a powerful software tool that provides the running time of (almost) any LWE attack.

```

.....: from estimator import *
.....: sig=3.19
.....:
.....: Xs=ND.UniformMod(3)
.....: Xe=ND.DiscreteGaussian(sig)
.....: n = 2^12
.....: q =2**135
.....:
.....: params = LWE.Parameters(n=n, q=q, Xs=Xs, Xe=Xe)
.....:
.....: LWE.primal_usvp(params)
.....: LWE.primal_bdd(params)
.....: LWE.dual(params)
rop: ≈2^100.7, red: ≈2^100.7, δ: 1.005647, β: 237, d: 7830, tag: usvp
rop: ≈2^100.2, red: ≈2^100.2, svp: ≈2^95.5, β: 235, η: 265, d: 8057, tag: bdd
rop: ≈2^101.9, mem: ≈2^53.2, m: ≈2^12.0, β: 237, d: 8190, v: 1, tag: dual

```

[1] Albrecht, Player, Scott - *On the concrete hardness of learning with errors* (2015)

How are FHE Parameters Selected Today?

Parameters for FHE scheme: $\lambda, n, q, \sigma_s, \sigma_e$

	Target parameter	Flexible	Easy to integrate with existing FHE libraries	Fast
Lattice Estimator [1]	λ	✓	✗	✗

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How are FHE Parameters Selected Today?

2. Precomputed tables. The Lattice Estimator has been adopted in [2] and [3] to provide specific tables for FHE parameters.

Homomorphic Encryption Standard

n	$\log_2(q)$	$\log_2(\sigma)$ (Classical)			distribution	n	security level	$\log q$	uSVP	dec	dual
		Binary	Ternary	Gaussian							
$\lambda = 128$											
630	32	17.9	16.6	14.2	uniform	1024	128	29	131.2	145.9	161.0
1024		7.6	6.3	4.5		192	21	192.5	225.3	247.2	
≥ 2048		2.0	2.0	2.0		256	16	265.8	332.6	356.7	
630	64	49.9	48.6	46.2		2048	128	56	129.8	137.9	148.2
750		46.8	45.5	43.0		192	39	197.6	217.5	233.7	
870		43.7	42.4	39.9		256	31	258.6	294.3	314.5	
1024		39.6	38.3	36.1		4096	128	111	128.2	132.0	139.5
2048		12.6	11.4	9.4		192	77	194.7	205.5	216.4	
≥ 4096		2.0	2.0	2.0		256	60	260.4	280.4	295.1	
630	64	49.9	48.6	46.2		8192	128	220	128.5	130.1	136.3
750		46.8	45.5	43.0		192	154	192.2	197.5	205.3	
870		43.7	42.4	39.9		256	120	256.5	267.3	277.5	
1024		39.6	38.3	36.1	16384	128	440	128.1	129.0	133.9	
2048		12.6	11.4	9.4	192	307	192.1	194.7	201.0		
≥ 4096		2.0	2.0	2.0	256	239	256.6	261.6	269.3		
630	64	49.9	48.6	46.2	32768	128	880	128.8	129.1	133.6	
750		46.8	45.5	43.0	192	612	193.0	193.9	198.2		
870		43.7	42.4	39.9	256	478	256.4	258.8	265.1		
1024		39.6	38.3	36.1							
2048		12.6	11.4	9.4							
≥ 4096		2.0	2.0	2.0							

Update version

[2] Albrecht et al. – *Homomorphic Encryption Security Standard*, HomomorphicEncryption.org, (2018)

[3] Bossuat, et al. - *Security Guidelines for Implementing Homomorphic Encryption* (2024)

How are FHE Parameters Selected Today?

Parameters for FHE scheme: $\lambda, n, q, \sigma_s, \sigma_e$

	Target parameters	Flexible	Easy to integrate with existing FHE libraries	Fast
Lattice Estimator [1]	λ	✓	✗	✗
Precomputed Tables [2,3]	λ, n, q, σ_e	✗	✓	✓

[1] Albrecht, Player, Scott - *On the concrete hardness of learning with errors* (2015)

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Lattice Estimator [1]	λ	✓	✗	✗
Precomputed Tables [2,3]	λ, n, q, σ_e	✗	✓	✓
Formula-based [4,5]	λ, n, q, σ_e	✓	✓	✓

[1] Albrecht, Player, Scott - *On the concrete hardness of learning with errors* (2015)

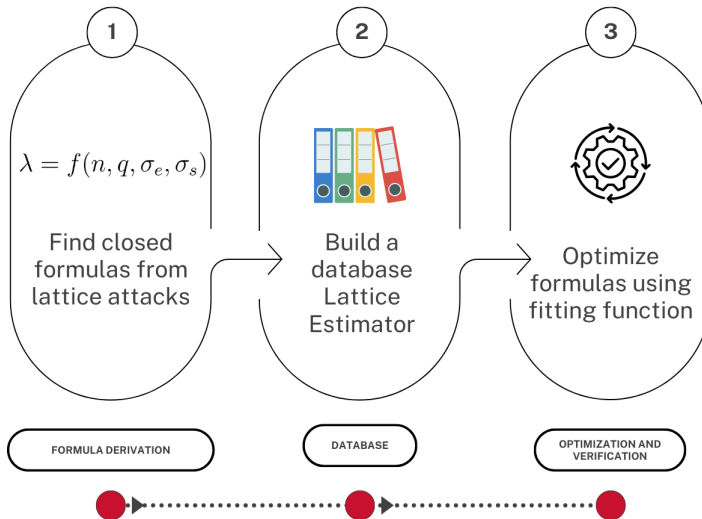
[2] Albrecht, et al. - *Homomorphic Encryption Security Standard* (2018)

[3] Bossuat, et al. - *Security Guidelines for Implementing Homomorphic Encryption* (2024)

[4] Kirshanova, Marcolla, Rovira - *Guidance for Efficient Selection of Secure Parameters for Fully Homomorphic Encryption* (2024)

[5] Biasioli, Kirshanova, Marcolla, Rovira - *A Tool for Fast and Secure LWE Parameter Selection: the FHE case* (2025)

Third Approach – a Formula-based Approach



Optimization – uSVP

- 1 We find the *optimal* β for the uSVP attack (where $\zeta = \sigma_e/\sigma_s$):

$$\beta = \frac{2n \ln(q/\zeta) \ln \left(\frac{n \ln(n/\ln q)}{2\pi e \ln(q/\sigma_e)} \right)}{\ln^2 \left(\frac{q \sqrt{n \ln(n/\ln(q/\sigma_e))} / \ln q}{2\pi e \sigma_e} \right)}$$

- 2 Given β , we obtain an expression for λ :

$$\lambda = 0.292\beta + \log_2 \left(8 \sqrt{\frac{2n \ln(q/\zeta)\beta}{\ln(\beta/(2\pi e))}} \right) + 16.4$$

Optimization – uSVP

- 1 We find the *optimal* β for the uSVP attack.
- 2 Given β , we obtain an expression for λ .
- 3 We **built** our **database**.

χ_s	Range of n	Range of $\log q$	σ_e	# points
$\mathcal{U}_3$	$[2^{10}, 2^{15}]$	$[10, 1600]$	3.19	5282
$\mathcal{U}_2$	$[2^{10}, 2^{11}]$	$[20, 64]$	3.19	42962

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For **BGV/BFV/CKKS** :

- ★ The **secret s** follows

Ternary distribution $\chi_s = \mathcal{U}_3$: Uniform over the ternary set $\{\pm 1, 0\}$, thus $\sigma_s = \sqrt{2/3}$.

Sparse distribution $\chi_s = \mathcal{HWT}(h)$ chooses a vector uniformly at random from $\{0, \pm 1\}^n$ with exactly h nonzero entries, where $h \leq n$ positive integer, thus $\sigma_s = \sqrt{h/n}$.

- ★ The **error e** follows the **discrete Gaussian distribution** $\chi_e = \mathcal{DG}(0, \sigma_e^2)$ with $\sigma_e = 3.19$.
- ★ The **LWE dimension $n \in \{2^{10}, \dots, 2^{16}\}$** and **modulus $\log q \in \{20, \dots, 2900\}$** depending on n and λ :
e.g. for $80 \leq \lambda \leq 192$ and $n = 2^{10} \implies 20 \leq \log q \leq 45$ or $n = 2^{16} \implies 1230 \leq \log q \leq 2900$.

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For **DM/CGGI** :

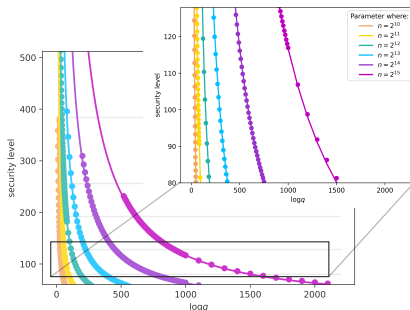
- ★ The **secret s** follows **binary** distribution $\chi_s = \mathcal{U}_2$: Uniform over the set $\{1, 0\}$, thus $\sigma_s = 1/2$.
- ★ The **error e** follows the **discrete Gaussian distribution** $\chi_e = \mathcal{DG}(0, \sigma_e^2)$, the centered at 0 with standard deviation σ_e that **varies**.
- ★ The **LWE dimension** $n \in \{2^{10}, 2^{11}\}$ and **modulus** $q \in \{2^{32}, 2^{64}\}$.

Optimization – uSVP

- 1 We find the *optimal* β for the uSVP attack.
- 2 Given β , we obtain an expression for λ .
- 3 We built our database.
- 4 We model our **approximation formula** with coupled optimization, finding λ

$$\lambda = A\beta + B \ln \left(\frac{2n \ln(q/\zeta)\beta}{\ln(\beta/(2\pi e))} \right) + C,$$

where A, B, C are fitted once, separately for $\chi_s = \mathcal{U}_2$ and $\chi_s = \mathcal{U}_3$.

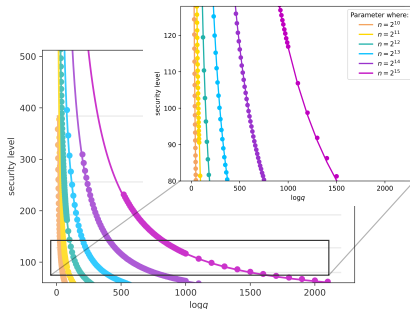


Optimization – uSVP

- 1 We find the *optimal* β for the uSVP attack.
- 2 Given β , we obtain an expression for λ .
- 3 We built our database.
- 4 We model our approximation formula with coupled optimization, finding
- 5 We **approximate** our formula **lambda**:

$$\lambda \approx \tilde{A} \ln \left(\frac{\tilde{B}n}{\ln q} \right) \frac{n}{\ln q} + \tilde{C} \ln n + \tilde{D}$$

Again four constants, fitted once per secret distribution: the formula is then a one-line computation.



Security formula for $\chi_s = \mathcal{U}_3$ and $\sigma_e = 3.19$

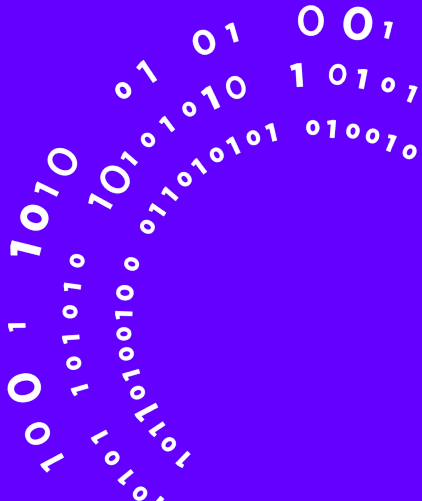
$n = 2^{10}$			
$\log q$	Estimator	lambda	lambda_s
27	129	130	128
28	124	125	123
30	115	116	114
32	107	108	106
43	78	78	78
48	70	70	70

$n = 2^{16}$			
$\log q$	Estimator	lambda	lambda_s
1776	134	128	127
1229	191	192	191
955	256	255	255

$n = 2^{15}$			
$\log q$	Estimator	lambda	lambda_s
810	140	140	140
881	128	128	128
930	120	121	120
1050	106	107	106
1200	92	93	92
1400	83	80	79

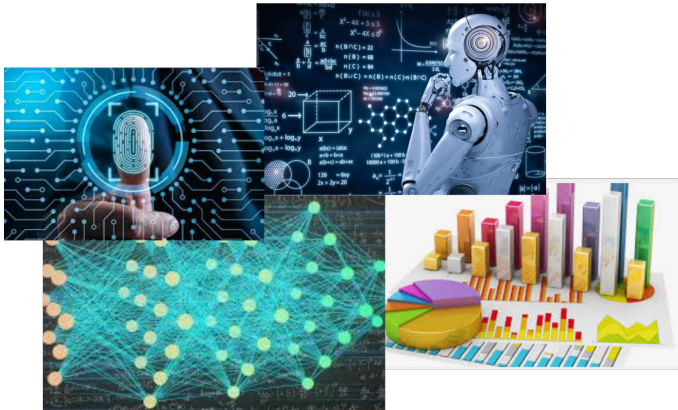
$n = 2^{17}$			
$\log q$	Estimator	lambda	lambda_s
3576	127	129	127
2469	192	192	190
1918	256	256	254

What it is used for

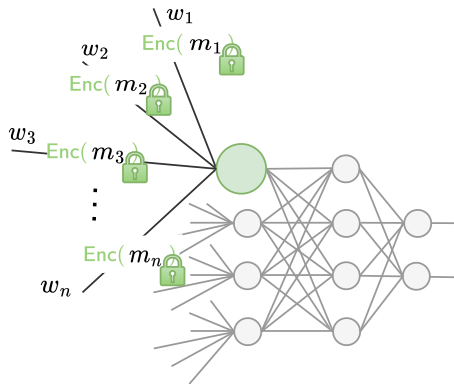


Some possible applications

- Medical
- Genomics
- Machine learning
- Statistics over sensitive data
- Private information retrieval
- Private set intersection
- Electronic voting



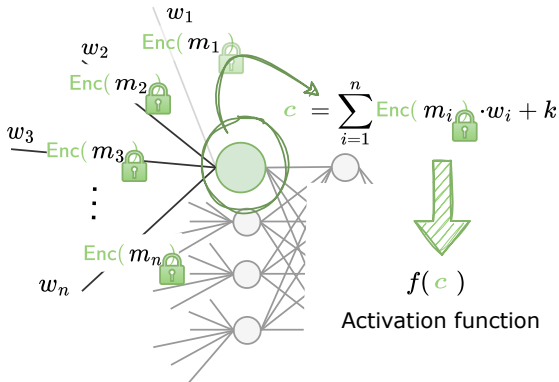
Neural Networks



The **activation function** used in Neural Networks is typically **non-arithmetic** function. For example

- ReLu function $f(x) = \max(0, x)$
- sign function $sgn(x)$
- hyperbolic tangent function $\tanh(x)$

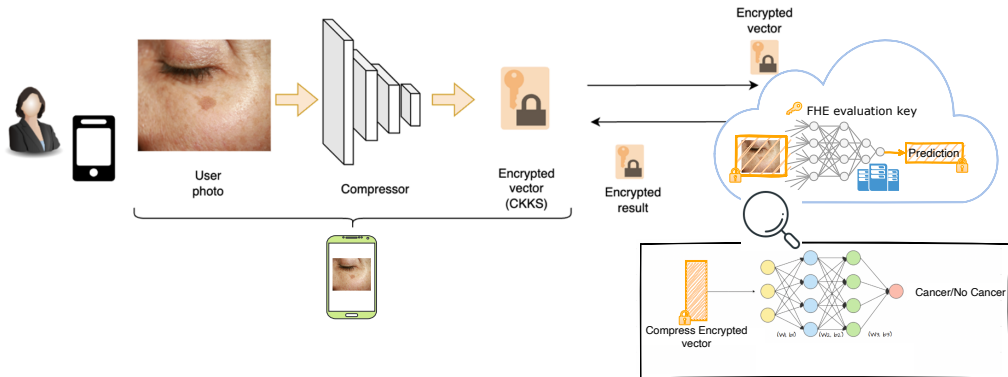
Neural Networks



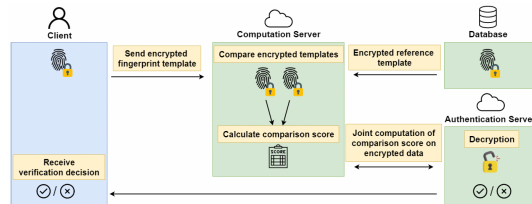
The **activation function** used in Neural Networks is typically **non-arithmetic**.

- 2° and 4° generation schemes supports linear functions $\Rightarrow$ we approximate the activation function, e.g. by a Chebyshev polynomial.
- 3° and 5° generation schemes use programmable bootstrapping

Privacy-preserving machine learning: skin cancer detection



Biometric authentication: two honest data points



✓ Iris [1], scheme BFV

features are **fixed-length binary vectors**, they fit the slots and the accuracy is preserved

✗ Fingerprint [2], scheme CKKS

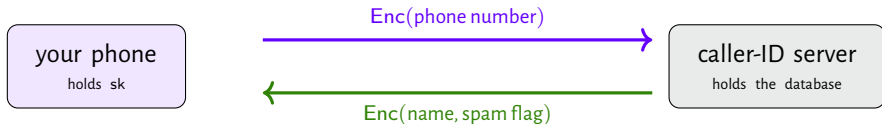
minutiae-based representations, one verification takes about **192 minutes**, not usable in practice

[1] Morampudi, Prasad, Raju. *Privacy-preserving iris authentication using fully homomorphic encryption* (2020).

[2] Bauspieß, Vad, Myrekrok, Costache, Kolberg, Rathgeb, Busch. *On the feasibility of fully homomorphic encryption of minutiae-based fingerprint representations* (2023).

Apple: private caller identification

An unknown number calls you. A caller-ID service can tell you who it is, but only if you **send it the number**: to learn who is calling, you hand over exactly the metadata you wanted to protect.

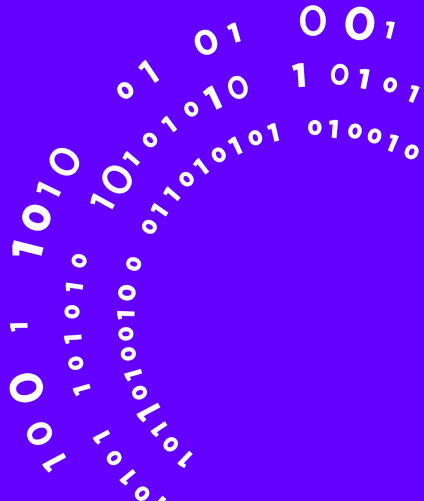


Live Caller ID Lookup [1], in iOS 18 and used by third-party providers: the number is encrypted **on the device**, the server answers with a homomorphic computation, and it learns **neither the number nor the answer**.

Scheme: BFV. 128-bit security. The implementation is open source.

[1] Apple, *Combining machine learning and homomorphic encryption in the Apple ecosystem* (2024), and the `swift-homomorphic-encryption` library.

Wrapping up



What to take home

- 1 Two bounds, opposite directions. Correctness pushes q **up**, while lattice attacks push it **down**. The dimension n is the dial that gives us room for both — but every extra bit comes at a cost.
- 2 Worst-case is clean; average-case is tight. Norms compose nicely, which makes worst-case analysis robust but often pessimistic. Modeling the actual noise distribution gives much tighter estimates — but requires more care.
- 3 There is no “magic” parameter set. Choosing FHE parameters means balancing correctness, security, and efficiency. Fortunately, we have tools to help us navigate this trade-off.

Yesterday: why FHE works at all.

Today: how to choose parameters that make it correct, secure, and practical.

**...Thank you for your
attention!**

